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ODD END SEMESTER EXAMINATION- 2024

Name of the Course: B. Tech (CSE) / B.Tech

Semester: III
Paper Code: TCS346

Time: 3 Hours

Maximum Marks: 100

Note:-

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20 marks)	CO5
(a)	The Health Department visited the government school to monitor the health of children; they recorded the following information: Name, Gender, Height (in meters), and Weight (in Kg). During the process, some values of Gender and Weight were missing. Develop the Python code to perform the following task using the Pandas library - Assume the data is available in health.csv, load the data into Data Frame - Replace the Gender missing values using the mode method and weight missing values with the mean method, update new values in the same Data Frame - Add the new column BMI into the Data Frame, and calculate the BMI value of the child based on a formula $BMI = \frac{Weight}{Height^2}$ - Categorize Children Based on BMI and create new Data Frames such as Under-Weight, Normal-Weight, Over-Weight, and Obesity based on the following conditions: - Under-Weight: BMI is below 18.5 - Normal-Weight: BMI is 18.5-24.9 - Over-Weight: BMI is 25.0-29.9 - Obesity: BMI is 30.0 and above	
(b)	Compare the following - Pandas vs Numpy - Matplotlib vs Seaborn	

(c) The GDP (Gross Domestic Product) contributions of the top six economies in 2024 are recorded as follows (in trillion USD):

Country	United States	China	Japan	Germany	India	United Kingdom
GDP (in Trillion USD)	25.5	18.7	5.1	4.2	3.9	3.2

Develop a Python code using the Matplotlib library to create a pie chart based on the below specification to visualize the GDP contributions of these six countries.

- Add labels to the pie chart corresponding to the country names.
- Start the first wedge at 90 degrees.
- Pull the wedge for the country with the highest GDP by a distance of 0.1 from the center of the pie.
- Use custom colors for each wedge.
- Add a legend to the chart for clarity.
- Enable shadow for the pie chart.

Q2	(20 marks)																																			
(a)	Discuss the different approaches in data cleaning with an example.																																			
(b)	Define correlation and explain its significance in data analysis. Differentiate between positive, negative, and zero correlation with appropriate examples to illustrate each type.																																			
(c)	The details of the employees working in a reputed company as given below (assume the data is available in the employee.csv file) <table><tr><th>Employee ID</th><th>Gender</th><th>Age</th><th>Experience (Years)</th><th>Salary (LPA)</th></tr><tr><td>101</td><td>Male</td><td>25</td><td>3</td><td>5</td></tr><tr><td>102</td><td>Female</td><td>28</td><td>5</td><td>6</td></tr><tr><td>103</td><td>Male</td><td>35</td><td>10</td><td>15</td></tr><tr><td>104</td><td>Female</td><td>32</td><td>8</td><td>10</td></tr><tr><td>105</td><td>Male</td><td>45</td><td>20</td><td>20</td></tr><tr><td>106</td><td>Female</td><td>30</td><td>6</td><td>8</td></tr></table> Develop a Python code to answer the following with appropriate title and axis labels - Plot histogram for the Age distribution of employees - Create scatter plots to analyze the following relationship between Experience and Salary - Generate box plots to explore the variability and outliers in the Salary grouped by Gender	Employee ID	Gender	Age	Experience (Years)	Salary (LPA)	101	Male	25	3	5	102	Female	28	5	6	103	Male	35	10	15	104	Female	32	8	10	105	Male	45	20	20	106	Female	30	6	8
Employee ID	Gender	Age	Experience (Years)	Salary (LPA)																																
101	Male	25	3	5																																
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103	Male	35	10	15																																
104	Female	32	8	10																																
105	Male	45	20	20																																
106	Female	30	6	8																																

CO6,
CO5

Q3	(20 marks)		
(a)	Discuss different types of Inheritance in Python with an example		
(b)	Compare the following concepts with the program example in Python i. Overloading vs Overriding ii. Private vs Public access-modifiers		CO4
(c)	Create a Python class called Smart Watch with Brand, Model Number, Features, and Price as variables within it. Develop a Python program to create N Smart Watch objects and print the Brand, Model Number, Features, and Price of these objects with suitable headings		
Q4	(20 marks)		
(a)	Elaborate the Pandas merging and grouping methods with examples		
(b)	A courier company charges for parcel delivery based on the weight of the parcels. The charges are as follows: • Parcels weighing 1 to 5 kg: Rs. 50 per kg. • Parcels weighing 6 to 15 kg: Rs. 40 per kg. • Parcels weighing 16 to 30 kg: Rs. 30 per kg. • Parcels weighing more than 30 kg: Rs. 20 per kg. Two regular customers, Ganesh and Subramanya, use the courier service frequently. Recently, they shipped parcels with the following weights (in kg): • Ganesh: [3, 12, 25, 7, 18, 34, 4] • Subramanya: [20, 5, 15, 33, 10] Develop a Python program to calculate and print the total delivery charges for Ganesh and Subramanya's shipments.		CO1, CO2
(c)	Discuss the File handling methods such as Create, Read, Write, Append and Delete with Python code snippets		
Q5	(20 marks)		
(a)	Discuss the following arguments in a Python function with code snippets i. Positional arguments ii. Keyword arguments iii. Variable-length positional arguments iv. Variable-length keyword arguments		CO3, CO2

(b)	Explain the Python count-controlled for loop and collection-controlled for loop. Develop a Python program using count-controlled for loop to calculate the sum of even numbers from 1 to 50 and collection-controlled for loop to iterate over the list ["apple", "banana", "cherry"] and print each fruit in uppercase	
(c)	The names of popular programming languages are stored in a Python tuple data structure. The names of the languages are stored in a variable named languages ('Python', 'Java', 'C++', 'Ruby', 'JavaScript', 'Go'). Develop a menu-driven Python program for the following operations on the Python tuple data structure: i. To add a new programming language to the tuple. ii. To update the name of a programming language in the tuple. iii. To remove a programming language from the tuple. iv. To display all programming language names.	